Aman Kumar

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Professional Summary

Full-Stack & **AI Engineer** with **2.5+ years** of experience building **AI-powered developer platforms, scalable code-generation systems**, and **real-time collaborative applications**. Experienced in **React**, **TypeScript**, **Node.js**, **Java**, **LangChain**, and **LangGraph**, with expertise in **LLM workflow orchestration, distributed systems, offline-first synchronization**, and **production deployment**. Proven track record of owning features end-to-end, shipping production applications, and leading engineering teams.

Technical Skills

AI & LLM Engineering: LangChain, LangGraph, LLM Workflow Orchestration, AI Code Generation, Agentic Workflows, Prompt Engineering
Languages & Core: JavaScript/TypeScript, Python, Java, C/C++, SQL
Frontend: React, HTML5, CSS3, Tailwind CSS, Bootstrap
Backend & Real-time: Node.js, Express.js, REST APIs, WebSockets, JWT
State & UI Systems: Zustand, React Flow, React DnD, AG Grid
Systems & Architecture: Distributed systems, Offline-first sync, Conflict detection & resolution, Multi-user collaboration, Low-code codegen, Workflow orchestration, Live preview, Version control
Data & DevOps: MongoDB, MySQL, Docker, GitHub, GitHub Actions, CI/CD

Professional Experience

Software Developer (Full-Time) July 2024 – Present
CTRI-Digital Governance, IIIT Bangalore
Owned end-to-end platform frontend — code generation, live preview, distributed collaboration, and AI codegen integration

- Engineered the **frontend generation module from scratch**, converting abstract UI models into production-ready **React applications** with complex component hierarchies, dynamic state management, and automatic routing
- Designed a full **code-generation pipeline** supporting dynamic component composition, style integration, and API bindings — enabling **non-technical users** to generate complete full-stack applications without writing a single line of code
- Developed a **distributed multi-user collaboration system** with offline-first sync, unit-level versioning, and hash-based conflict detection — supporting concurrent development across **10+ simultaneous users**
- Architected AI-powered frontend and backend **code-generation pipelines** using **LangChain** and **LangGraph**, orchestrating **multi-step LLM workflows** for **planning**, **code generation**, **validation**, and **full-stack application development**.
- Spearheaded a **drag-and-drop UI builder** using React, TypeScript, React DnD, and Zustand with real-time component manipulation and live state preview
- Implemented **live preview** functionality enabling real-time UI rendering without full application generation, improving developer iteration speed by **60%**
- Delivered a production grade **No-Dues application** for IIIT Bangalore **(nodues.iiitb.net) end-to-end in 4 days** using RASP — from requirements and architecture design to full-stack code generation (React + AI-generated Spring Boot backend), manual testing, and deployment — serving **300+ students** across departments
- Deployed and managed **3+ production-grade services** (RASP core, Collaboration Server, AI Codegen, Workflow Service, IR-to-Code Service) using **Docker** — handling containerization, environment config, and multi-service orchestration
- Mentored **20+ interns and junior engineers**, leading code reviews and architectural decisions across multiple teams

Software Developer (Intern) Jan 2024 – June 2024
CTRI-Digital Governance, IIIT Bangalore
Built foundational frontend generation module and component pipeline for a model-driven low-code platform

- Engineered the core **frontend generation engine** using React.js, TypeScript, and Node.js, translating metadata-driven UI configurations into production-ready React components with integrated styling and state management
- Established **Node.js APIs** to convert abstract UI models into downloadable React application files, supporting **5+ component types** with dynamic prop binding and style injection
- Collaborated with senior engineers to architect the **component generation pipeline**, establishing extensible patterns adopted across the full platform

Education

M.Tech, Computer Science & Engineering July 2022 – July 2024
International Institute of Information Technology, Bangalore

Achievements & Exhibitions

- Exhibitor, **Bangalore International Tech Summit 2025** — Represented CTRI-DG @ IIIT Bangalore and demonstrated the full scope of RASP to industry leaders, government officials, and international delegates; live demos covered the drag-and-drop UI builder, model-driven workflow engine, full-stack application generation and the complete end-to-end developer experience
- **GATE 2022** — Secured **97.96 percentile** in Computer Science and Engineering